

The Life Cycle of Stars

A beginner's tour of how stars are born, live, and die —
explained from the very beginning, no prior knowledge needed.

First Things First: What Even *Is* a Star?

A star is a **huge, glowing ball of hot gas** held together by its own gravity — and powered by a process called **fusion**, where tiny atoms of hydrogen squeeze together to form helium, releasing enormous amounts of light and heat.



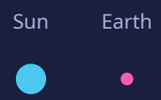
What it looks like

A dot of light in the night sky. Up close: a giant fireball. Our Sun is a star — the closest one to Earth.



Why it shines

Not burning like wood — it **fuses** atoms in its core. That reaction shines for millions or billions of years.



How big?

The Sun fits ~1.3 million Earths inside it. Some stars are hundreds of times bigger than the Sun.

The Big Picture: Every Star Lives 3 Phases

Think of it like a life story: **born**, a long stable **adulthood**, and eventually a **death**. The details of the ending depend on one thing: the star's **mass** (how much stuff it's made of).

1

🌱 Birth

A star starts as a cold, spread-out cloud of gas and dust floating in space. Gravity pulls it together until it lights up.

2

⚖️ Main Sequence (Adulthood)

The star settles into a stable routine, calmly turning hydrogen into helium. This is its longest, calmest phase.

3

🍂 Death

When the fuel starts running out, the star swells up and changes dramatically. How it ends depends on its mass.

⭐ **Key word to remember:** a **nebula** is the giant cloud of gas and dust where every star is born.

Phase 1 — Birth: How a Star Lights Up



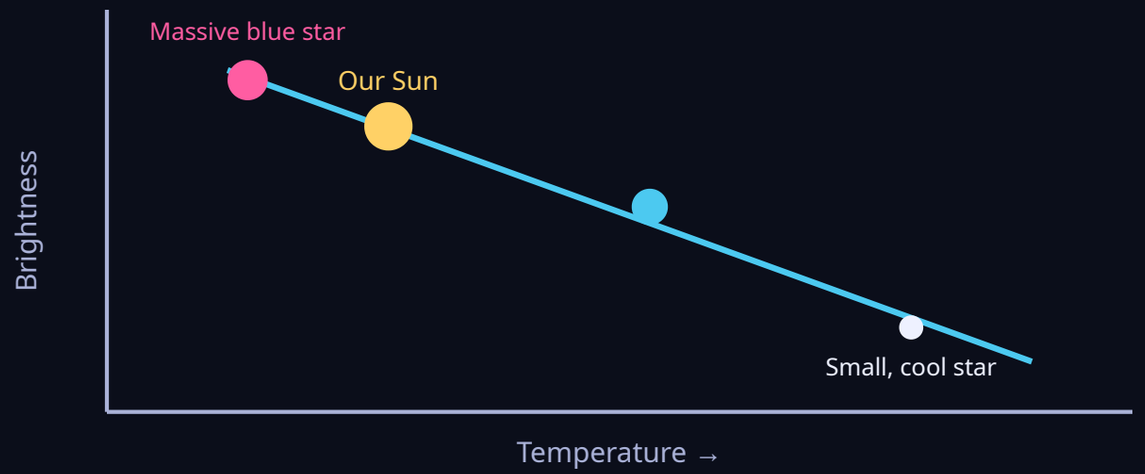
1. A nebula (gas cloud) 2. Squeezed into a ball 3. Fusion sta

Step 1: In space there are enormous clouds of gas and dust called **nebulae** (say: NEB-you-lee).

Step 2: Gravity — the same force that keeps you on the ground — slowly pulls the cloud inward, squishing it into a tight ball that gets hotter and hotter.

Step 3: When the center reaches about 10 million °C, hydrogen atoms start fusing. The ball ignites and becomes a **star**.

Phase 2 — Main Sequence: The Calm Middle of Life



What happens here: the star spends this entire phase calmly fusing hydrogen into helium in its core.

It's a balancing act: gravity pulls the star inward, while the energy from fusion pushes outward. These two forces balance perfectly, so the star stays the same size.

How long does it last? Our Sun will spend about **10 billion years** in this phase. Bigger stars are greedier — they burn through their fuel in just a few million years.

Phase 3 — Death: Running Out of Fuel

Eventually, the hydrogen in the core runs low. The balance breaks, and the star begins to change shape — **swelling up into a giant**. How big it swells depends on how massive it was:

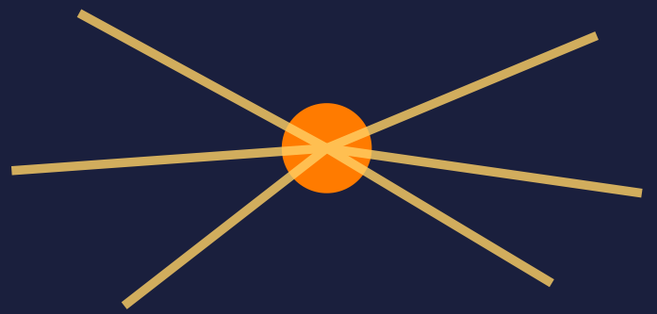
SMALLER STAR (LIKE OUR SUN)



Red Giant

The star puffs up to many times its old size and turns reddish. Picture the Sun growing so large it swallows Mercury and Venus.

MUCH BIGGER STAR



Red Supergiant

A massive star swells even bigger and starts fusing heavier elements — all the way up to iron, which is where trouble begins.

From here, the story splits into **two possible paths** — determined by the star's mass.

Path A — The Gentle Ending (stars like our Sun)

Stars with less than about **8 times the Sun's mass** don't go out with a bang. They fade away quietly, over an incredibly long time.

Main Sequence
Stable Sun-like star

→

Red Giant
Swollen and red

→

Planetary Nebula
Outer layers gently drift off into space

→

White Dwarf
Hot, Earth-sized leftover core

→

Black Dwarf
A cold, dark ember (takes trillions of years)



Red giant | Planetary nebula | White dwarf

No explosion — the star sheds its outer layers like a cloak, leaving only its hot core behind, which slowly cools down over trillions of years.

Path B — The Explosive Ending (massive stars)

Stars with **more than about 8 times the Sun's mass** end their lives in the most violent event in the universe: a **supernova** — an explosion so bright it can briefly outshine an entire galaxy.

Main Sequence
Bright blue giant

→

Red Supergiant
Fuses elements up to iron

→

Supernova
The core collapses and the star explodes

→

Two possible leftovers:



Neutron Star

If the leftover core is small enough, it becomes a neutron star — the whole core squeezed into a ball only ~20 km wide (city-sized!), spinning super fast.



Black Hole

If the core is too heavy, gravity wins completely and it collapses into a black hole — a region of space so dense that nothing, not even light, can escape it.

Summary: The Whole Story in One Glance

Nebula
cloud of gas & dust

→

Protostar
gravity squeezes it

→

Main Sequence
stable star (adulthood)

→

Red Giant / Supergiant
fuel runs low, star swells

Path A → Quiet Fade

Planetary nebula → **White dwarf** → Black dwarf
(for stars like our Sun)

Path B → Explosive End

Supernova → **Neutron star** or **Black hole**
(for massive stars)

💡 One last wonder: the iron in your blood and the calcium in your bones were forged inside stars and scattered by supernovae. **You are made of stardust.**